Sandeep Poloju

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Summary

Software Engineer with 3+ years of end-to-end ownership shipping production-ready distributed systems in fast-paced, Agile environments. Skilled across the full stack, building modern frontends with Angular/React, scalable backends with Spring Boot and FastAPI (Python), and intelligent features powered by LLMs, RAG pipelines, and Agentic AI. Brings a strong foundation in system design and DevOps (AWS, Kubernetes), with a track record of delivering 10X performance gains.

Experience

Infosys

Software Engineer –II *Oct 2022 – Dec 2023*

- Optimized the search suggestion API for a high-traffic FinTech platform by re-architecting the core Cypher query and implementing Protobuf, reduced latency by **90% (10X)** for a vastly improved user experience.
- Accelerated a financial services platform's AI capabilities by integrating LLMs to boost NLU search accuracy from **85% to 92%** and designing a Graph Neural Network (GNN) microservice for link prediction.
- Designed a dynamic multi-tenant routing layer using **NGINX** reverse proxy and regex capture groups. Resolved the need for an intermediate routing microservice, reducing infrastructure complexity and request overhead.
- Automated microservice deployments on a Rancher cluster using Docker and Kubernetes to ensure scalability.
- **Led a team of three in an Agile/Scrum environment** to automate the regression testing pipeline, eliminating 20 hours of manual effort per release.
- Tech Stack: **Python, Java, PyTorch, LLMs, NGINX, Docker, Kubernetes, Rancher, Protractor**

Software Engineer –I *May 2021 – Oct 2022*

- Simplified a microservice's data serialization to create a minimal API payload, cutting network traffic by **68% (465KB to 148KB)** and minimizing network latency.
- Engineered a Python translation layer within a core microservice to convert GraphQL queries directly to Cypher, by passing an inefficient intermediate service and improving system performance.
- **Established RESTful APIs** for the User Preference using Node.js and PostgreSQL, enabling persistent storage of user settings and personalized dashboard configurations.
- Refactored a legacy Angular front-end **using SOLID principles** and decoupled the Graph Visualization module by implementing the Adapter Design Pattern, establishing a scalable codebase.
- Standardized accessibility compliance across the Angular front-end by implementing WCAG 2.1 Level AA patterns; validated 100% parity for screen readers (NVDA/VoiceOver) and keyboard navigation, resolving 40+ UI accessibility barriers.
- Tech Stack: **Angular, TypeScript, Python, Node.js, PostgreSQL, Neo4j, Cypher, GraphQL, RESTful APIs, Spring Boot**

University of Maryland Apr 2024 – Jan 2025

Research Assistant (Software Engineer) *College Park, MD*

- Developed a full-stack AgriTech web application to predict Brown Patch disease severity, assembling a **10GB+ training dataset** by unifying data from PDFs, websites, and NOAA weather data from **AWS S3**.
- Improved model accuracy **from 74% to 91%** through Köppen climate-based regional segmentation and stepwise feature selection; trained Random Forest and Neural Network models using **AWS SageMaker**.
- Deployed serverless inference via **AWS Lambda and API Gateway** for cost-efficient scaling.
- Tech Stack: **Python, TesseractOCR, BeautifulSoup, REST APIs, AWS (S3, Lambda, API Gateway), Tensorflow, Scikit-learn**

Smart Interviews Feb 2021 – May 2021

Software Developer & Instructor

- Developed and optimized interactive features for the React-based SmartInterviews platform, improving UI responsiveness and reducing load times.
- Mentored 160 students as a teaching assistant in the fields of **data structures, algorithms**, and problem-solving.
- Tech Stack: **React, JavaScript, HTML5, CSS3**

Projects

Student Q&A RAG System | *FastAPI, LangGraph, MCP, ChromaDB, Docker, Ollama* | GitHub — Feb 2026

- Built a RAG pipeline over academic PDFs using **LangGraph** for stateful orchestration, which classifies, retrieves, and rejects off-topic queries via conditional routing.
- Designed a provider-agnostic LLM layer switchable between cloud (Nebius) and local Ollama inference for air gap deployment; containerized with Docker for one-command portable deployment.
- Integrated a **Model Context Protocol (MCP)** server for dynamic web search via DuckDuckGo, enabling LangGraph to route off-topic queries to live internet results instead of rejecting them.

GREBoost | *Angular, TypeScript, Flask, MongoDB, Vercel, GitHub Actions* | GitHub — Sep – Nov 2024

- Developed a microservices-based e-learning platform, implementing RESTful APIs and an Angular frontend, deployed on Vercel with automated CI/CD using GitHub Actions.

Education

University of Maryland, College Park — College Park, MD
Master of Engineering, Software Engineering — GPA: 3.93 — *Jan 2024 – Dec 2025*

Certifications

- **Claude Code in Action** by Anthropic | Credential

Achievements & Publications

- **Top 100 Rank (Top 0.06%)**, HackwithInfy National Coding Competition (167,000+ participants).
- **Runner-up among 95 teams** in the WatsonX Chatbot Challenge | Submission
- Competitive programmer: Highest rating CodeChef 1799, CodeForces 1370.
- **Solved 600+** advanced algorithmic problems across major platforms (LeetCode, CodeForces, CodeChef, HackerRank).
- Published: "Analyzing Twitter Event Hype vs. Reality" – NLP sentiment analysis (IJRAR) | Paper

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Cypher (Graph Query), Shell Scripting
AI/ML: RAG, LangChain, LangGraph, OpenAI/Gemini APIs, Cursor, Claude Code, PyTorch, TensorFlow, NLP, MCP
Frontend: Angular (Expert), React, Flutter, HTML5, SCSS, RxJS
Backend: Spring Boot, FastAPI, Flask, GraphQL, gRPC
DevOps: AWS (Lambda, SageMaker, EC2), Docker, Kubernetes, NGINX, GitHub Actions, Vercel, CI/CD
Databases: PostgreSQL, MongoDB, Neo4j, Redis